

FUNZIONI RAZIONALI

$$f(x) = \frac{N(x)}{D(x)} \quad \begin{array}{l} \nwarrow \text{POLINOMI} \\ \swarrow \end{array}$$

OGGI:

$$\text{GRADO } D(x) = 2 \quad \begin{array}{l} \swarrow \Delta > 0 \\ \Delta = 0 \\ \searrow \Delta < 0 \end{array}$$

• CASO I: $\Delta > 0$

es)

$$\int \frac{x-3}{2x^2-x-1} dx$$

$$\text{GRADO } N(x) = 1 < \text{GRADO } D(x) = 2$$

$$\text{NO DIVISIONE} \quad \frac{N(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$$

$$D(x) = 2x^2 - x - 1$$

$$\Delta = 1 + 4 \cdot 2 = 9 > 0$$

CERCHIAMO GLI ZERI DI $D(x)$

$$x = \frac{1 \pm \sqrt{\Delta}}{4} = \frac{1 \pm 3}{4} \begin{matrix} -\frac{1}{2} \\ 1 \end{matrix}$$

$$D(x) = 2\left(x + \frac{1}{2}\right)(x-1) = (2x+1)(x-1)$$

$$\frac{x-3}{2x^2-x-1} = \frac{x-3}{(2x+1)(x-1)}$$

$$= \frac{A}{2x+1} + \frac{B}{x-1}$$

$$= \frac{A(x-1) + B(2x+1)}{(2x+1)(x-1)}$$

$$x-3 = Ax - A + 2Bx + B$$

$$1 \cdot x - 3 = (A+2B)x + (B-A)$$

$$\begin{cases} A+2B=1 \\ B-A=-3 \end{cases} \quad \begin{matrix} 3B=-2 & B=-\frac{2}{3} \\ A=1-2B=1+\frac{4}{3}=\frac{7}{3} \end{matrix}$$

$$\int f(x) dx = \int \frac{\frac{7}{3}}{2x+1} + \frac{-\frac{2}{3}}{x-1} dx$$

$$= \frac{7}{3} \int \frac{2x}{2x+1} dx - \frac{2}{3} \int \frac{1}{x-1} dx$$

$$= \frac{7}{3} \frac{\log|2x+1|}{2} - \frac{2}{3} \log|x-1| + C$$

$$\log(2x+1)' = \frac{1}{2x+1} \cdot (2x+1)'$$

$$= \frac{2}{2x+1}$$

• CASO III : $\Delta < 0$

es)

$$\int \frac{x+1}{9x^2+4} dx$$

GRADO $N(x)=1 <$ GRADO $D(x)=2$

→ NO DIVISIONE

$$\Delta = 0 - 4 \cdot 9 \cdot 4 < 0$$

$\Rightarrow 9x^2+4$ NON SI PUO' SCOMPORRE

$$9x^2+4 \neq (\dots) \cdot (\dots)$$

$$\int \frac{x+1}{9x^2+4} dx = \int \frac{x}{9x^2+4} + \frac{1}{9x^2+4} dx$$

$$= \underbrace{\int \frac{x}{9x^2+4} dx}_a + \underbrace{\int \frac{1}{9x^2+4} dx}_b$$

$$a) \int \frac{x}{9x^2+4} dx =$$

$$y = 9x^2+4$$

$$\frac{dy}{18} = 18x dx$$

$$= \int \frac{1}{y} \frac{dy}{18}$$

$$= \frac{1}{18} \int \frac{1}{y} dy = \frac{1}{18} \log|y| + C$$

$$= \frac{1}{18} \log|9x^2+4| + C$$

↓
> 0

$$= \frac{1}{18} \log(9x^2+4) + C$$

$$b) \int \frac{1}{9x^2+4} dx$$

|| SICCOME NON C'É LA "X" A NUMERATORE, LA SOSTITUZIONE $y = 9x^2+4$ NON AIUTA

$$\frac{1}{9x^2+4} = \frac{1}{4\left(\frac{9}{4}x^2+1\right)}$$

$$\int \frac{1}{9x^2+4} dx = \int \frac{1}{4} \frac{1}{\left(\frac{3}{2}x\right)^2+1} dx$$

$$= \frac{1}{4} \int \frac{1}{\left(\frac{3}{2}x\right)^2+1} dx \quad \frac{3}{2}x = y$$

$$\frac{3}{2}dx = dy$$

$$= \frac{1}{4} \int \frac{1}{y^2+1} \frac{2}{3} dy \quad dx = \frac{2}{3} dy$$

$$= \frac{1}{4} \frac{2}{3} \int \frac{1}{y^2+1} dy$$

$$= \frac{1}{6} \arctan y + C$$

$$= \frac{1}{6} \arctan\left(\frac{3}{2}x\right) + C$$

INFINE:

$$\int dx = \frac{1}{18} \log(9x^2+4) + \frac{1}{6} \arctan\left(\frac{3}{2}x\right) + C$$

$$\textcircled{a}) \int \frac{1}{x^2+x+1} dx$$